

Introduction into Possibilistic Logic

Possibilistic Logic - An Overview by Didier Dubois and Henri Prade

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- ▶ **Definition:** Possibilistic logic is a weighted logic that handles uncertainty in a qualitative way by adding certainty to classical logic formulas
- ▶ **Applications:** Used in expert systems, decision-making, and AI for reasoning under uncertainty.

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What are Possibility Distributions?

- ▶ A **possibility distribution** is a mapping:

$$\pi : U \rightarrow S,$$

where:

- ▶ U : Set of states
- ▶ S : Totally ordered scale ($S = [0, 1]$ or finite chains).
- ▶ $\pi(u)$: Degree of possibility (plausibility) of state u .
- ▶ $\pi(u) = 0$: State u is **impossible**.
- ▶ $\pi(u) = 1$: State u is **fully possible**.

Normalization Condition

At least one state $u \in U$ must have $\pi(u) = 1$ (an actual state exists).

- ▶ A distribution π is **at least as specific** as π' if:

$$\pi(u) \leq \pi'(u), \quad \forall u \in U.$$

- ▶ Examples:

- ▶ **Complete knowledge:** $\pi(u_0) = 1$, $\pi(u) = 0$ for all $u \neq u_0$.

- ▶ **Complete ignorance:** $\pi(u) = 1$ for all $u \in U$.

- ▶ Principle of minimal specificity: It states that any hypothesis not known to be impossible cannot be ruled out. It is a minimal commitment, cautious, information principle.

- Possibility measure:

$$\Pi(A) = \sup_{u \in A} \pi(u).$$

Evaluates the extent to which A is consistent with π .

- Necessity measure:

$$N(A) = \inf_{u \notin A} (1 - \pi(u)).$$

Evaluates the extent to which A is certainly implied by π .

Duality Relationship

$$N(A) = 1 - \Pi(A^c).$$

Where A^c is the complement of A .

► **Possibility: Maxitivity Property**

$$\Pi(A \cup B) = \max(\Pi(A), \Pi(B)).$$

► **Necessity: Minimization Property**

$$N(A \cap B) = \min(N(A), N(B)).$$

- ▶ Certainty-qualified information: "A is certain to degree α " is modeled by the constraint:

$$N(A) \geq \alpha$$

- ▶ Least Specific Possibility Distribution:

$$\pi_{(A,\alpha)}(u) = \begin{cases} 1 & \text{if } u \in A, \\ 1 - \alpha & \text{otherwise.} \end{cases}$$

- ▶ If $\alpha = 1$: $\pi(A, \alpha)$ becomes the characteristic function of A .
- ▶ If $\alpha = 0$: $\pi(A, \alpha)$ represents total ignorance.

Example

- ▶ A doctor is trying to diagnose a patient based on symptoms and diseases:
 - ▶ $U = \{\text{flu}, \text{cold}, \text{allergy}\}$.
 - ▶ Possibility distribution π based on symptom severity:

$$\pi(\text{flu}) = 0.9, \quad \pi(\text{cold}) = 0.6, \quad \pi(\text{allergy}) = 0.3.$$

- ▶ The doctor evaluates the possibility and necessity measures for a hypothesis A :

$$A = \{\text{flu}, \text{cold}\}$$

- ▶ Calculation:

$$\Pi(A) = \sup_{u \in A} \pi(u) = \max(\pi(\text{flu}), \pi(\text{cold})) = 0.9.$$

$$N(A) = \inf_{u \notin A} (1 - \pi(u)) = 1 - \pi(\text{allergy}) = 1 - 0.3 = 0.7.$$

- ▶ Interpretation:
 - ▶ Possibility ($\Pi(A)$): Flu or cold is highly possible (0.9).
 - ▶ Necessity ($N(A)$): It is moderately certain (0.7) that the patient has either flu or cold.

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- ▶ A basic possibilistic logic formula is represented as:

$$(a, \alpha)$$

where:

- ▶ a : A classical logic formula (e.g., $p \rightarrow q$).
- ▶ $\alpha \in [0, 1]$: Certainty level (minimum necessity degree of a).
- ▶ ****Interpretation****:
 - ▶ (a, α) : "Formula a is at least certain to degree α ."
 - ▶ The higher α , the stronger the belief in a .

- ▶ ****Weakened Certainty****:

$$(a, \alpha) \Rightarrow (a, \beta), \quad \text{if } \alpha \geq \beta$$

- ▶ ****Min-Based Conjunction****:

$$N(a \wedge b) = \min(N(a), N(b))$$

- ▶ ****Resolution****:

$$(\neg a \vee b, \alpha), (a \vee c, \beta) \Rightarrow (b \vee c, \min(\alpha, \beta))$$

- ▶ ****Weakened Certainty****:

$$(a, \alpha) \Rightarrow (a, \beta), \quad \text{if } \alpha \geq \beta$$

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Example (Continued)

Using rules:

$$(\neg \text{fever} \vee \text{flu}, 0.9), \quad (\text{fever} \vee \text{headache}, 0.8)$$

Inference:

$$(\text{flu} \vee \text{headache}, 0.8)$$

- ▶ Possibilistic logic can handle conflicting information by:
 - ▶ Identifying an **inconsistency level** ($inc - l$).
 - ▶ Filtering out rules with certainty levels below $inc - l$.
- ▶ **Example**:

$$\{(a, 0.8), (\neg a, 0.7)\}$$

- ▶ Inconsistency level: $inc - l = 0.7$.
- ▶ Filtered result: $(a, 0.8)$ dominates.

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What is Possibilistic Logic?

- ▶ ****Definition****: Possibilistic logic is an extension of classical logic that incorporates a certainty level for each formula.
- ▶ ****Purpose****: Designed for reasoning under qualitative uncertainty and preferences.
- ▶ ****Core Characteristics****:
 - ▶ A formula in possibilistic logic is a pair (a, α) , where:
 - ▶ a : A classical logical formula (propositional or first-order logic).
 - ▶ $\alpha \in [0, 1]$: A certainty level.
 - ▶ Certainty levels (α) represent the confidence or priority of the formula.
- ▶ ****Scope****:
 - ▶ Classical logic is retained: Formulas are either true or false.
 - ▶ Extensions exist for fuzzy predicates, but basic possibilistic logic does not handle degrees of truth.

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- ▶ A possibilistic logic formula is written as:

$$(a, \alpha), \quad \text{where } \alpha \in [0, 1].$$

- ▶ Interpretation:

- ▶ a : A propositional formula.
- ▶ α : The minimum necessity level associated with a .

- ▶ ****Examples****:

- ▶ $(p \rightarrow q, 0.8)$: "If p , then q with certainty 0.8."
- ▶ $(\neg p, 1)$: "Not p is absolutely certain."

- ▶ A possibilistic logic formula is written as:

$$(a, \alpha), \quad \text{where } \alpha \in [0, 1].$$

- ▶ Interpretation:

- ▶ a : A propositional formula.
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- ▶ ****Examples****:

- ▶ $(p \rightarrow q, 0.8)$: "If p , then q with certainty 0.8."
- ▶ $(\neg p, 1)$: "Not p is absolutely certain."

Key Idea

Possibilistic logic extends classical logic by associating a priority (certainty) level to each formula while maintaining Boolean semantics.

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- ▶ ****Certainty Weakening****:

$$\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta).$$

- ▶ ****Modus Ponens****:

$$(\neg a \vee b, \alpha), \quad (a, \alpha) \vdash (b, \alpha).$$

- ▶ ****Resolution****:

$$(\neg a \vee b, \alpha), \quad (a \vee c, \alpha) \vdash (b \vee c, \alpha).$$

- ▶ ****Weakest Link Resolution****:

$$(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta)).$$

- ****Certainty Weakening****:

$$\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta).$$

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- ****Weakest Link Resolution****:

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Example

Consider:

$$\begin{aligned} &(\neg \text{fever} \vee \text{flu}, 0.9), \\ &(\text{fever} \vee \text{headache}, 0.8). \end{aligned}$$

► ****Definition**:**

If $a \vdash b$ in classical logic, then $(a, \alpha) \vdash (b, \alpha)$.

► Derived from resolution:

$$(\neg a \vee b, 1), \quad (a, \alpha) \vdash (b, \alpha).$$

► Example:

► $a \rightarrow b$: $p \wedge q \vdash p$.

► With certainty: $(p \wedge q, 0.7) \vdash (p, 0.7)$.

► ****Definition**:**

If $a \vdash b$ in classical logic, then $(a, \alpha) \vdash (b, \alpha)$.

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► Example:

► $a \rightarrow b$: $p \wedge q \vdash p$.

► With certainty: $(p \wedge q, 0.7) \vdash (p, 0.7)$.

Key Property

Any valid deduction in propositional logic is valid in possibilistic logic, with certainty levels propagated.

- ▶ A possibilistic logic base is a collection of weighted formulas:

$$\{(a_1, \alpha_1), (a_2, \alpha_2), \dots\}.$$

- ▶ Clauses can be rewritten in terms of their weights:

- ▶ Example:

$$(a \wedge b, \alpha) \quad \Leftrightarrow \quad \{(a, \alpha), (b, \alpha)\}.$$

- ▶ Logical tautologies (t) hold with any certainty level:

$$(t, \alpha), \quad \forall \alpha \in [0, 1].$$

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► Rule base:

$(\neg \text{fever} \vee \text{flu}, 0.9),$
 $(\neg \text{cough} \vee \text{cold}, 0.8),$
 $(\text{fever} \vee \text{headache}, 0.7).$

► Facts:

$(\text{fever}, 1), \quad (\text{cough}, 1).$

► Inference:

► From Modus Ponens:

$(\text{flu}, 0.9), \quad (\text{cold}, 0.8).$

► From Resolution:

$(\text{flu} \vee \text{headache}, 0.7).$

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- ▶ Possibilistic logic provides a robust framework for reasoning under uncertainty.
- ▶ Key features:
 - ▶ Certainty levels for qualitative inference.
 - ▶ Compatibility with classical logic.
 - ▶ Handling of conflicting information.
- ▶ Applications:
 - ▶ Medical diagnosis, decision-making, and knowledge representation.

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What is Possibilistic Logic?

- ▶ **Definition**: A logical framework for reasoning under uncertainty.
- ▶ **Purpose**: Handles incomplete information and default rules with exceptions.
- ▶ **Core Idea**:
 - ▶ Classical logic enriched with certainty levels.
 - ▶ Focus on **default reasoning**: "What typically happens."
- ▶ **Applications**:
 - ▶ Nonmonotonic reasoning.
 - ▶ Knowledge representation and reasoning about exceptions.

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- ▶ ****Definition****: A logical framework for reasoning under uncertainty.
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 - ▶ Knowledge representation and reasoning about exceptions.

Example Context

Default reasoning can handle statements like:

- ▶ "Birds fly (usually)." ($b \rightarrow f$).
- ▶ "Penguins do not fly." ($p \rightarrow \neg f$).
- ▶ "Penguins are birds." ($p \rightarrow b$).

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- ▶ A possibilistic formula is a pair:

$$(a, \alpha), \quad \text{where } \alpha \in [0, 1].$$

- ▶ Interpretation:

- ▶ a : A propositional formula.
- ▶ α : Certainty level (degree of belief in a).

- ▶ ****Examples****:

- ▶ $(b \rightarrow f, 0.9)$: Birds fly with certainty 0.9.
- ▶ $(p \rightarrow \neg f, 1)$: Penguins do not fly, absolutely certain.

- ▶ A possibilistic formula is a pair:

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Key Idea

Certainty levels enrich classical logic, enabling reasoning about typical cases and exceptions.

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- ▶ ****Certainty Weakening****:

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- ▶ ****Modus Ponens****:

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- ▶ ****Resolution****:

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- ▶ ****Weakest Link Resolution****:

$$\underline{(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta)).}$$

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- ▶ ****Weakest Link Resolution****:

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Penguin Example

Consider:

$$(\neg b \vee f, 0.9), \quad (\neg p \vee \neg f, 1), \quad (\neg p \vee b, 1).$$

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- ▶ Default rule:

$$\underline{\Pi(a \wedge b) > \Pi(a \wedge \neg b)}.$$

- $a \rightarrow b$: "If a , then b is more plausible than $\neg b$."

- ▶ In terms of necessity:

$$\underline{N(b|a) = 1 - \Pi(\neg b|a) > 0}.$$

- ▶ Comparative relation:

$$\underline{a \wedge b >_{\Pi} a \wedge \neg b}.$$

- ▶ Default rule:

$$\underline{\Pi(a \wedge b) > \Pi(a \wedge \neg b)}.$$

- $a \rightarrow b$: "If a , then b is more plausible than $\neg b$."

- ▶ In terms of necessity:

$$\underline{N(b|a) = 1 - \Pi(\neg b|a) > 0}.$$

- ▶ Comparative relation:

$$\underline{a \wedge b >_{\Pi} a \wedge \neg b}.$$

Penguin Example Constraints

$$b \wedge f >_{\Pi} b \wedge \neg f \quad (\text{birds fly}).$$

$$p \wedge \neg f >_{\Pi} p \wedge f \quad (\text{penguins don't fly}).$$

$$p \wedge b >_{\Pi} p \wedge \neg b \quad (\text{penguins are birds}).$$

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- ▶ ****Possibility Distribution****:

$\pi(\omega)$ values assigned to worlds ω .

- ▶ **Worlds**:

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\}.$$

- ▶ ****Possibility Distribution****:

$\pi(\omega)$ values assigned to worlds ω .

- ▶ **Worlds**:

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\}.$$

- ▶ ****Constraints****:

$$C_1 : \max(\pi(\omega_6), \pi(\omega_7)) > \max(\pi(\omega_4), \pi(\omega_5)).$$

$$C_2 : \max(\pi(\omega_5), \pi(\omega_1)) > \max(\pi(\omega_3), \pi(\omega_7)).$$

$$C_3 : \max(\pi(\omega_5), \pi(\omega_7)) > \max(\pi(\omega_1), \pi(\omega_3)).$$

- ▶ Factual knowledge:

$(b, 1)$ (Tweety is a bird).

- ▶ Possibilistic inference:

$K \cup \{(b, 1)\} \vdash (f, 1/3)$.

- Default rule: Tweety flies with certainty $1/3$.

- ▶ Adding $p = 1$ (Tweety is a penguin):

$K \cup F \vdash (\neg f, 2/3)$.

- New inference: Tweety does not fly.

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- ▶ Possibilistic logic effectively models:
 - ▶ Default reasoning with exceptions.
 - ▶ Reasoning under uncertainty.
- ▶ ****Penguin Example****:
 - ▶ Illustrated how rules like "birds fly" and "penguins don't fly" can be reconciled.
 - ▶ Demonstrated inference and conflict resolution.
- ▶ ****Applications****:
 - ▶ Knowledge bases.
 - ▶ Default rule modeling.
 - ▶ Nonmonotonic reasoning.

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- ▶ ****Medical Diagnosis****:
 - ▶ Rules with certainty for differential diagnoses.
- ▶ ****Decision-Making****:
 - ▶ Selecting actions under incomplete knowledge.
- ▶ ****Knowledge Bases****:
 - ▶ Handling updates with belief revision.

- ▶ ****Medical Diagnosis****:
 - ▶ Rules with certainty for differential diagnoses.
- ▶ ****Decision-Making****:
 - ▶ Selecting actions under incomplete knowledge.
- ▶ ****Knowledge Bases****:
 - ▶ Handling updates with belief revision.

Example Wrap-Up

Throughout the presentation, we used:

- ▶ Syntax: Representation of rules $((a, \alpha))$.
- ▶ Measures: Calculating $\Pi(A)$ and $N(A)$.
- ▶ Inference: Combining uncertain rules.
- ▶ Inconsistencies: Managing conflicts in rules.

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- ▶ ****Default Reasoning****: Handling incomplete knowledge in expert systems.
- ▶ ****Medical Diagnosis****: Prioritizing plausible diagnoses based on symptoms.
- ▶ ****Decision-Making****: Evaluating options under uncertainty.
- ▶ ****Fuzzy Control Systems****: Representing and processing vague information.

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Possibility distribution:

U is a set of affairs.

A possibility distribution is a mapping π from U to a totally ordered scale S , with top denoted by 1 and bottom by 0. In the finite case $S = \{1 = \lambda_1 > \dots \lambda_n > \lambda_{n+1} = 0\}$.

The function π represents the state of knowledge of an agent (about the actual state of affairs), also called an epistemic state.

$\pi(u) = 0$ means that state u is rejected as impossible;

$\pi(u) = 1$ means that state u is totally possible (= plausible).

Specificity: A possibility distribution π is said to be at least as specific as another π' if and only if for each state of affairs u : $\pi(u) \leq \pi'(u)$. Then, π is at least as restrictive and informative as π' , since it rules out at least as many states with at least as much strength.

Complete knowledge: for some u_0 , $\pi(u_0) = 1$ and $\pi(u) = 0$, $\forall u \neq u_0$ (only u_0 is possible).

Complete ignorance: $\pi(u) = 1, \forall u \in U$ (all states are possible).

Principle of minimal specificity: It states that any hypothesis not known to be impossible cannot be ruled out. It is a minimal commitment, cautious, information

principle.

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Block

This is a regular block.

- ▶ This is an item in a block.

Block

This is an alert block.

- ▶ This is an item in an alert block.

Block

This is an example block.

- ▶ This is an item in an example block.

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We can highlight text. There are also a darker option, and a yellow option.

The same options are also available in math mode: $a + b = c$

23 Appendix

This is an appendix.

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